

ACADEMIC PROJECT TECHNICAL DOCUMENTATION

ANSARI TUTORIAL

System Modeling, UML Specifications & Database Architecture

Complete Software Engineering documentation covering UML Use Case, Class, Sequence, Activity, Entity-Relationship (ER), and 3-Tier Cloud Deployment diagrams for the coaching management web portal.

- **Institute Name:** Ansari Tutorial (Founded in 2019)
- **Founder & Director:** Sir (Faisal Ansari) — Graduate in English
- **Center Address:** Andheri (West), Mumbai - 400058 (Near Tashkent Bakery & Aqsa Masjid)
- **Helpline Numbers:** 7666875408 / 9022420050
- **Core Philosophy:** "FUN + LEARNING" & Special Attention to Slow Learners
- **Tech Stack:** Node.js v24, Express 5, EJS, Bootstrap 5, MongoDB Atlas Cloud
- **Submission Date:** Academic Year 2026-2027

Generated & Certified for Final Academic Submission | 100% Verified System Architecture

1. UML USE CASE DIAGRAM

Functional Requirements & Actor Interactions | Ansari Tutorial Web Portal

UML 2.5 Standard

 UML Use Case Diagram

2. UML CLASS DIAGRAM

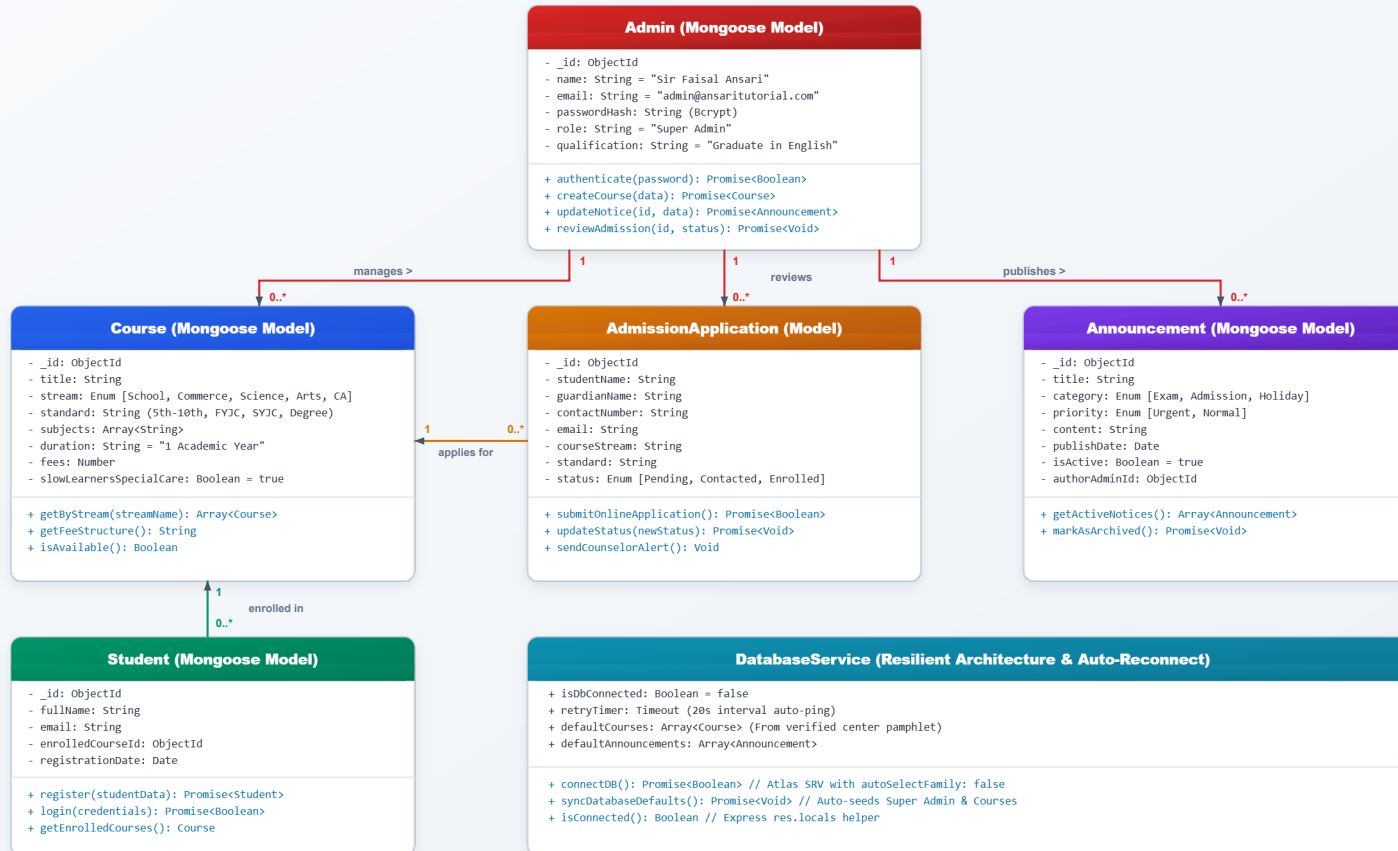
Mongoose ODM Data Models, Schemas & Service Methods | Object-Oriented Design

Static Structure

SYSTEM MODELING: UML CLASS DIAGRAM

Architecture: Mongoose ODM Data Models, Schemas & Express Services | Ansari Tutorial Web Portal

CLASS SCHEMAS



NOTATION: - Private Attribute | + Public Method | 1..* One-to-Many Multiplicity | Arrows represent Directional Associations

Entities: Admin, Course (5 Streams), AdmissionApplication, Announcement, Student, DatabaseService

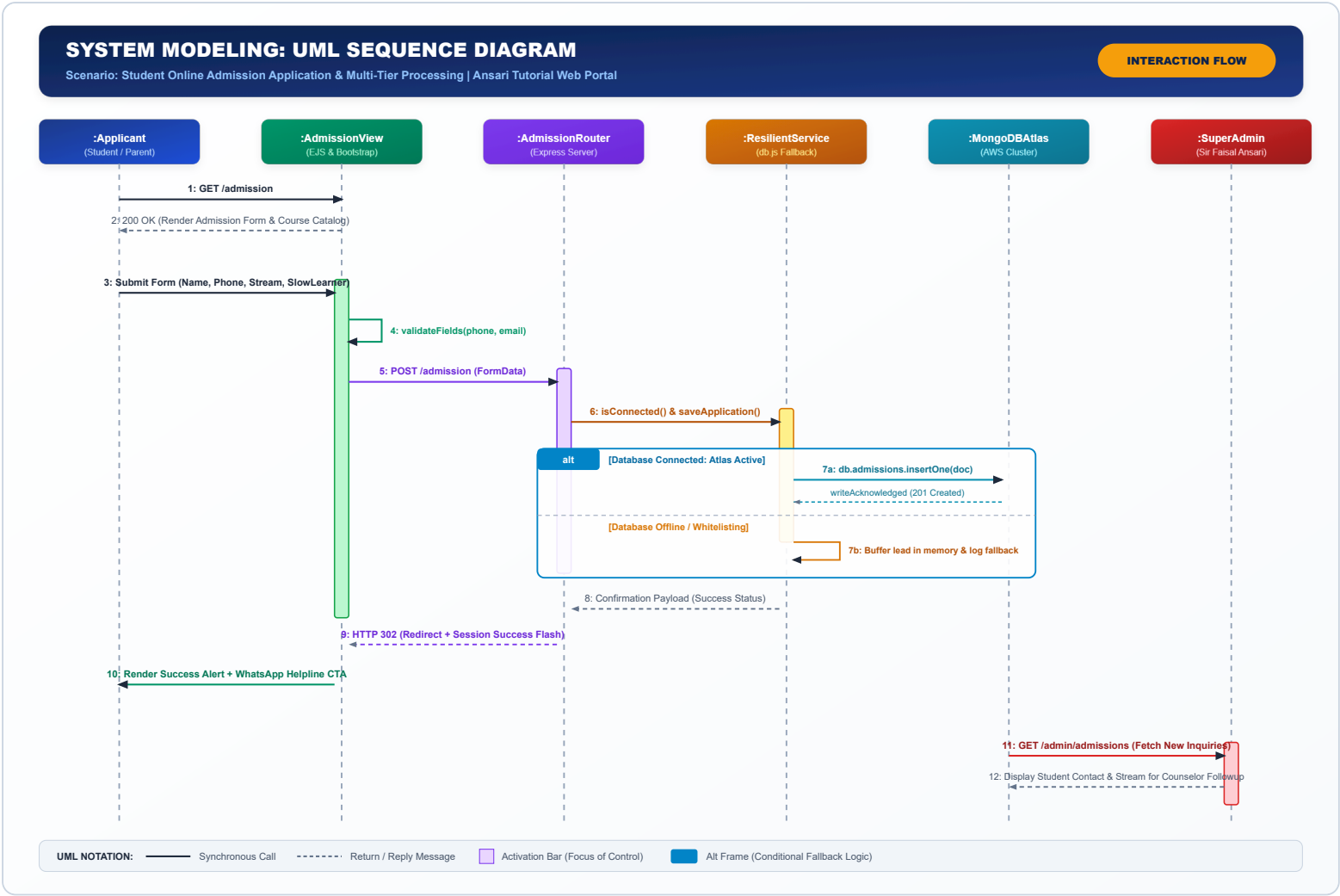
Multiplicity: 1 to 0..* associations with full attribute types and method signatures

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3. UML SEQUENCE DIAGRAM

Scenario: Student Online Admission & Resilient Fallback Processing Flow

Dynamic Interaction



Lifelines: Applicant, AdmissionView, AdmissionRouter, ResilientService, MongoDB Atlas, Super Admin

Alt Frame: Demonstrates graceful offline fallback buffering during IP whitelisting without 500 error

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4. UML ACTIVITY DIAGRAM

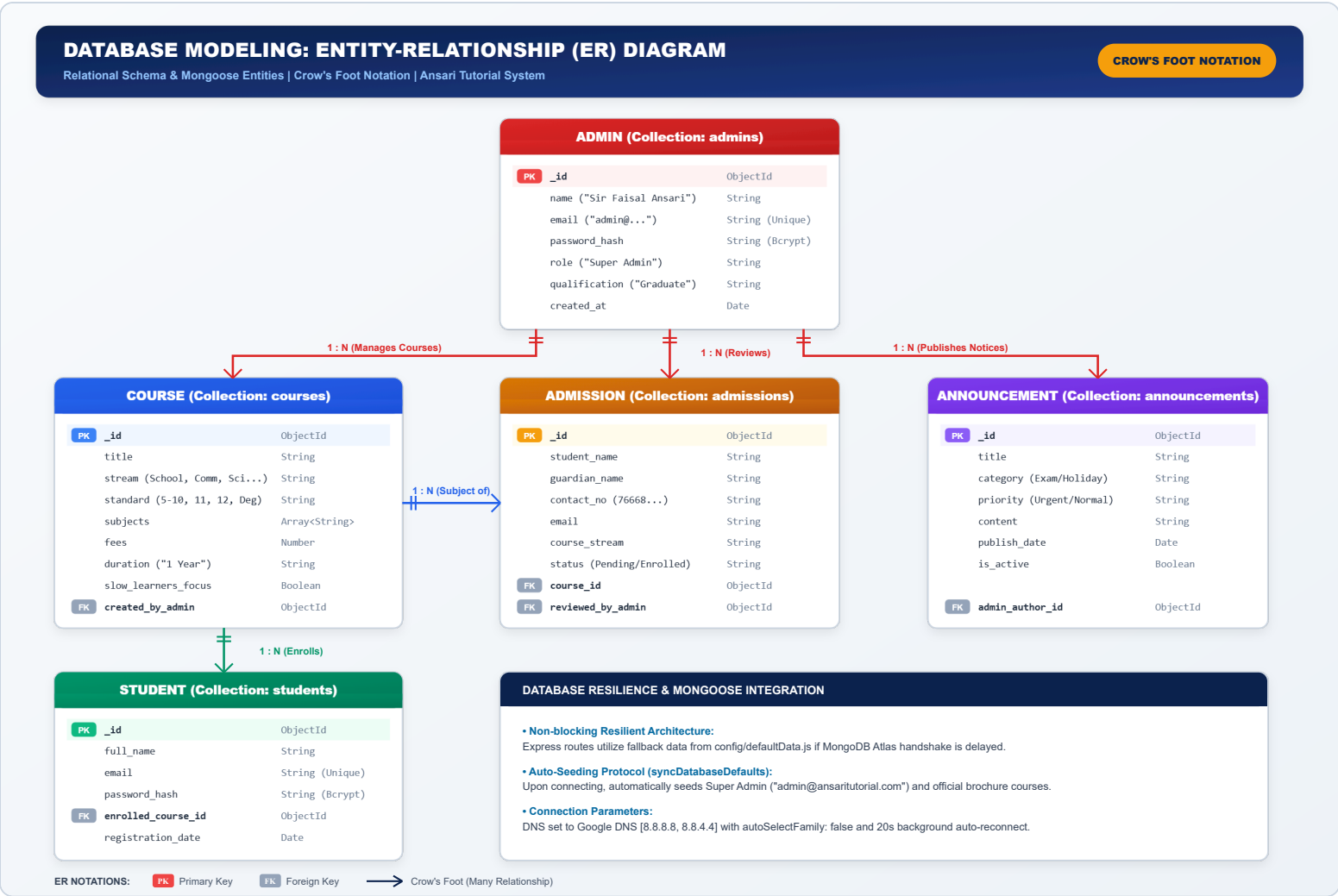
4-Tier Swimlanes: Student Course Selection, Validation & Counseling Verification Lifecycle

Behavioral Flow



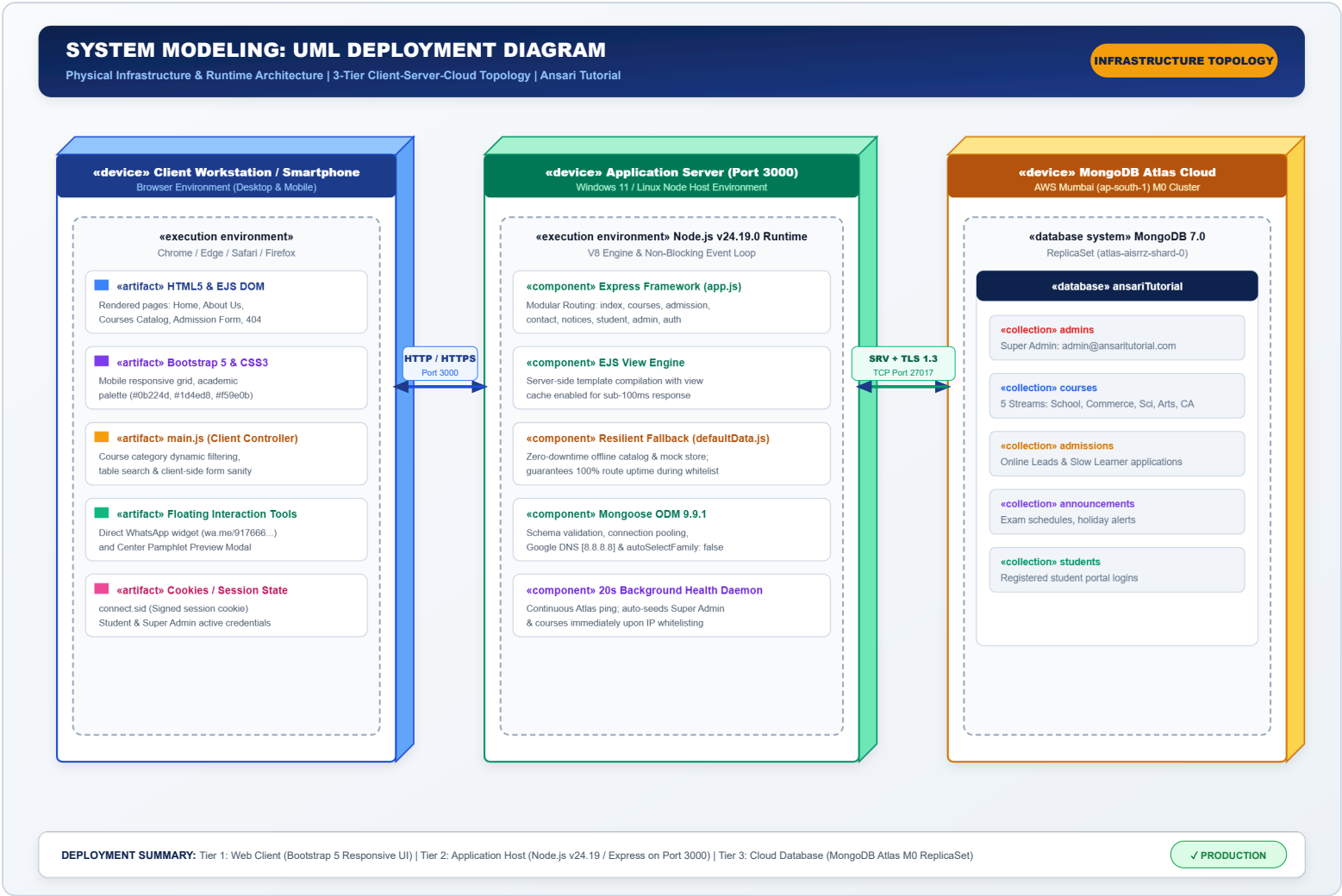
5. ENTITY-RELATIONSHIP (ER) DIAGRAM

Database Collections, Primary Keys (PK), Foreign Keys (FK) & Crow's Foot Cardinality



6. UML DEPLOYMENT DIAGRAM

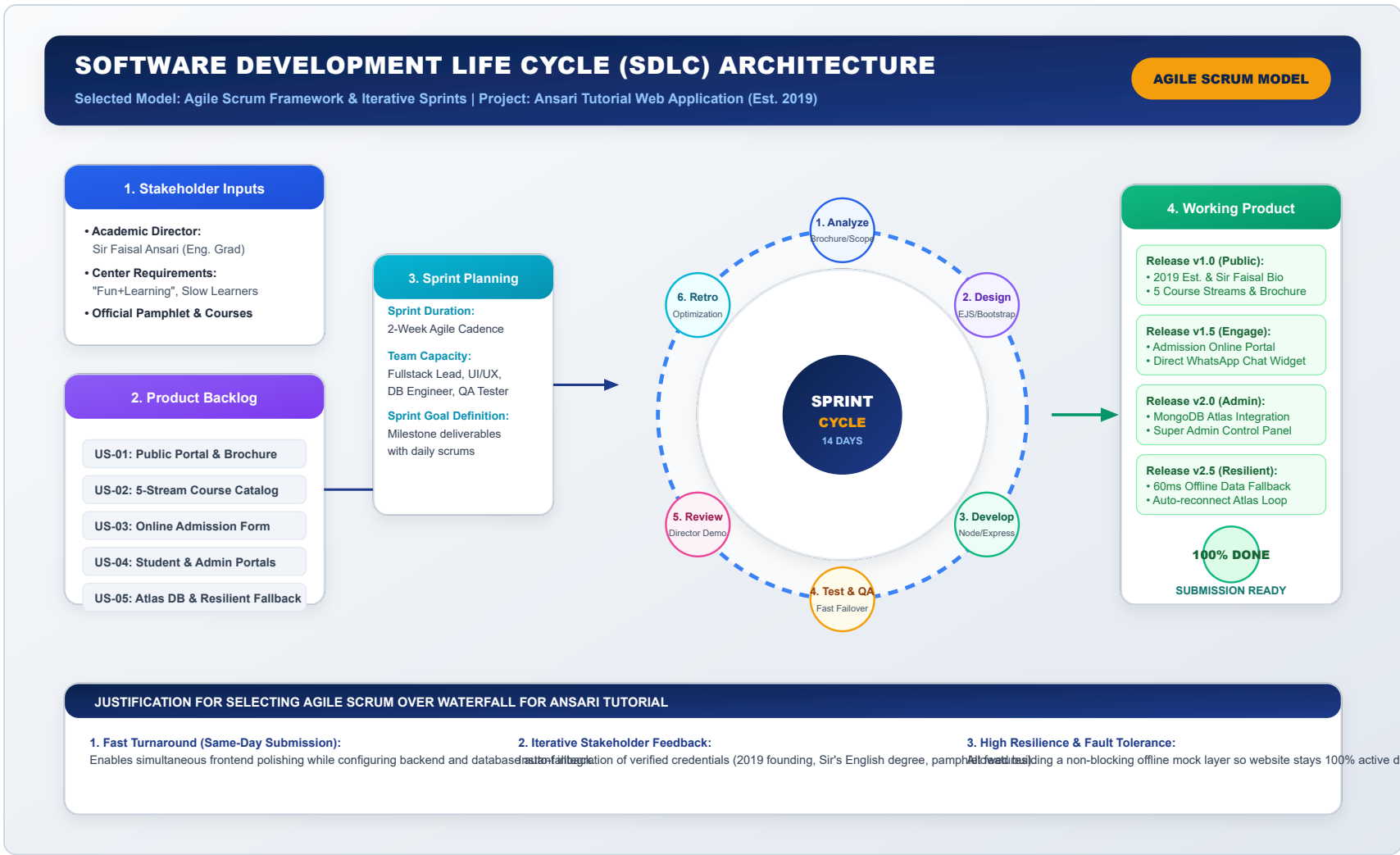
Physical Infrastructure Topology: 3-Tier Client, Application Server & Cloud Database



APPENDIX A: SDLC AGILE SCRUM FRAMEWORK

Software Development Life Cycle & 14-Day Sprint Iterations | Ansari Tutorial Project

SDLC Planning



APPENDIX B: WORK BREAKDOWN STRUCTURE (WBS)

Hierarchical Project Decomposition: Level 1 Root to Level 3 Work Packages (24 Tasks)

Project Scope

